

Desigualdad de Young para convoluciones

Proposición 1 (Young). Sean $f \in \mathcal{L}^1(\mathbb{R}^n)$, $g \in \mathcal{L}^p(\mathbb{R}^n)$ con $p \in [1, \infty]$

$$\implies \|f * g\|_p \leq \|f\|_1 \|g\|_p.$$

En particular, $|f * g(x)| < \infty$ c.t.p. $x \in \mathbb{R}^n$.

Demostración: Veámoslo en tres casos mutuamente excluyentes:

Caso I: Si $p = \infty$, por la convolucion/??, tenemos que $\|f * g\|_\infty \leq \|f\|_1 \|g\|_\infty$.

Caso II: Si $p = 1$, por el Teorema de Fubini, tenemos que

$$\begin{aligned} \|f * g\|_1 &= \int_{\mathbb{R}^n} |f * g(x)| \, dx = \int_{\mathbb{R}^n} \left| \int_{\mathbb{R}^n} f(x-y) g(y) \, dy \right| \, dx \\ &\leq \int_{\mathbb{R}^n} \int_{\mathbb{R}^n} |f(x-y)| |g(y)| \, dy \, dx \stackrel{\text{Fubini}}{=} \int_{\mathbb{R}^n} |g(y)| \left(\int_{\mathbb{R}^n} |f(x-y)| \, dx \right) \, dy = \\ &\leq \int_{\mathbb{R}^n} |g(y)| \|f\|_1 \, dy = \|f\|_1 \|g\|_1. \end{aligned}$$

Caso III: Si $p \in (1, \infty)$, tenemos que

$$\|f * g\|_p^p = \int_{\mathbb{R}^n} \left| \int_{\mathbb{R}^n} f(x-y) g(y) \, dy \right|^p \, dx \leq \int_{\mathbb{R}^n} \left(\int_{\mathbb{R}^n} |f(x-y)| |g(y)| \, dy \right)^p \, dx. \quad (1)$$

Por otro lado, tomando $q \in (1, \infty)$ como el exponente conjugado de p , tenemos que

$$\int_{\mathbb{R}^n} |f(x-y)| |g(y)| \, dy = \int_{\mathbb{R}^n} |f(x-y)|^{1/q} |f(x-y)|^{1/p} |g(y)| \, dy.$$

Entonces, por la desigualdad de Hölder para $|f(x-\cdot)|^{1/q}$ y $|f(x-\cdot)|^{1/p} |g|$, tenemos

$$\begin{aligned} \int_{\mathbb{R}^n} |f(x-y)| |g(y)| \, dy &\leq \left(\int_{\mathbb{R}^n} |f(x-y)| \, dy \right)^{\frac{1}{q}} \left(\int_{\mathbb{R}^n} |f(x-y)| |g(y)|^p \, dy \right)^{\frac{1}{p}} = \\ &\leq \|f\|_1^{1/q} \left(\int_{\mathbb{R}^n} |f(x-y)| |g(y)|^p \, dy \right)^{\frac{1}{p}}. \end{aligned}$$

Sustituyendo con esta expresión en (??), tenemos que

$$\begin{aligned}
\|f * g\|_p^p &\leq \int_{\mathbb{R}^n} \left(\|f\|_1^{1/q} \left(\int_{\mathbb{R}^n} |f(x-y)| |g(y)|^p dy \right)^{\frac{1}{p}} \right)^p dx = \\
&\leq \|f\|_1^{\frac{p}{q}} \int_{\mathbb{R}^n} \int_{\mathbb{R}^n} |f(x-y)| |g(y)|^p dy dx \stackrel{\text{Fubini}}{=} \|f\|_1^{\frac{p}{q}} \int_{\mathbb{R}^n} |g(y)|^p \left(\int_{\mathbb{R}^n} |f(x-y)| dx \right) dy = \\
&= \|f\|_1^{\frac{p}{q}} \|f\|_1 \int_{\mathbb{R}^n} |g(y)|^p dy = \|f\|_1^p \|g\|_p^p.
\end{aligned}$$

Por tanto, elevando a $1/p$, llegamos a $\|f * g\|_p \leq \|f\|_1 \|g\|_p$. ■

Referenciado en

- `lem-transformada-fourier-convolucion`

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